

Punjab University Lahore (2022)
Math (201)

Question #01

- i) All vector space must be abelian group under multiplication. Justify your answer with example.

In a vector space, the scalar multiplication is defined between a scalar and a vector, not between two vectors.

(e.g.) $V = \mathbb{R}^2$

$$\alpha(x, y) = (\alpha x, \alpha y) \quad \text{where } \alpha \text{ is a scalar } \mathbb{R}^2$$

$$(x, y) + (z, w) = (x + z, y + w)$$

$$(x, y) + (z, w) = (z, w) + (x, y)$$

Therefore a vector space is an abelian group under addition, but

DATE: _____

not under multiplication
ii) Show that any linearly independent set of vector of any vector space can be extended to a basis for V .

Let V be a vector space and let $S = \{v_1, v_2, v_3, \dots, v_n\}$ be a linearly independent set of vectors in V . We want to show that S can be extended to basis for V .

Step 1:- If S is already a spanning vector for V , then it is a basis for V .

Step 2:- If S is not a spanning set for V , then there exists a vector v in V that cannot be written as a linear combination of vector in S .

Step 3:- Let $S' = S \cup \{v\}$. We claim that S' is still linearly independent. Suppose $(a_1, a_2, \dots, a_n)$ and (b) .

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n + b v = 0$$

Therefore, any linear independly

set of vector in a vector space can be extended to basis for the vector space.

iii) let V be a vector space over any field and $W \subseteq V$. Then W is subspace of V or not?

let $V = \mathbb{R}^3$

$$W = \{ (x, y, z) \mid x^2 + y^2 + z^2 = 25 \} \subseteq \mathbb{R}^3$$

$$(4, 3, 0) \in W \subseteq \mathbb{R}^3$$

but $k = 2 \in \mathbb{R}$

$$\begin{aligned} k(4, 3, 0) &= 2(4, 3, 0) \\ &= (8, 6, 0) \end{aligned}$$

$$8^2 + 6^2 + 0^2 = 100 \neq 25$$

$$k(4, 3, 0) \notin W$$

So, W is not subspace of $\mathbb{R}^3$.

iv) Define:-

Orthogonal matrix:- Orthogonal matrix preserve the Euclidean norm and are widely used in linear algebra, statistics and many areas of science and

DATE: _____

DAY: _____

engineering.

$Q^T = Q^{-1}$ (the transpose of Q is inverse)

$Q^T Q = Q Q^T = I$ (identity matrix).

Similar matrix- Similar matrix

are matrices that have the same eigenvalues and can be transformed into each other through a series of row and column operations.

$$B = P^{-1} A P$$

v) **Define-**

Linear dependence- Linear dependence refers to the relationship between a set of vectors where one or more vectors can be expressed as a linear combination of the others.

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$$

Linear independence- Linear independence is a concept in linear algebra that describes

DATE: _____

DAY: _____

a set of vector that cannot be expressed as a linear combination of each vector.

$$\{v_1, v_2, \dots, v_n\}$$

$$\{a_1 v_1, a_2 v_2, \dots, a_n v_n\}$$

vi) **Definer:-**

determinants:- Determinants are scalar value that can be computed from the elements of a square matrix.

Cofactors & minors Cofactors are the element of a matrix that are used to compute the determinant of the matrix.

They are defined as the minor of the matrix multiplied by sign factor. $C_{ij} = (-1)^{(i+j)} \times M_{ij}$

Question #02

Long Questions:-

-1) let A and B be two matrix of order 6 such that $\det(AB)^2 = 72$

DATE: _____

DAY: _____

$\det(A^2 B^2) = 144$ Find ~~($\det(A)$)~~.
 $\det(A)$, $\det(2A)$ and
 $\det(AB^6)$

Example #5 page# 197

2) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined

$$T(x_1, x_2, x_3) = (x_1, -x_2, x_3)$$

Is T is linear Find (NT) and
 $R(T)$

Ex:- 6.4 Question #12

3) Find solution by Gauss-Jordan
Elimination method.

$$2x_1 - x_2 - x_3 = 4, \quad 3x_1 - 2x_2 + 4x_3 = 11,$$

$$3x_1 - 2x_2 + 4x_3 = 11$$

Example (4) Page (159)

4) Find eigen values and
corresponding Eigen vectors
of matrix $A = \begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix}$

DATE: _____

DAY: _____

- 1) Find characteristics eq. $\det(A - \lambda I) = 0$
- 2) Solve λ (eigen values)
- 3) Find eigen vectors $(A - \lambda I)x = 0$.

$$1) \det(A - \lambda I) = \det \begin{bmatrix} 5-\lambda & 9 & 3 \\ 3 & 5-\lambda & 6 \\ -1 & -5 & -3-\lambda \end{bmatrix}$$

$$= (5-\lambda)(5-\lambda)(3-\lambda) - 30 - 9(-9 - 3(3+\lambda)) - 3(-15 + 5(3+\lambda))$$

$$= (5-\lambda)(15 - 8\lambda + \lambda^2) + 81 + 27\lambda + 9\lambda^2 + 45 - 75\lambda + 15\lambda^2$$

$$= 15 - 8\lambda + \lambda^2 - 75\lambda + 81 + 27\lambda - 15\lambda^2 + 45 - 75\lambda + 15\lambda^2$$

$$= -30\lambda^2 - 121\lambda + 141$$

$$= 0$$

Solve λ (eigen values):-

$$-30\lambda^2 - 121\lambda + 141 = 0 \quad \text{divide by } -30$$

$$\lambda^2 + 4.033\lambda - 4.7 = 0$$

Solve the quadratic eq.:-

$$\lambda = -6.033, \quad \lambda = 0.767$$

Find the eigen vectors:-

$$\lambda = -6.033$$

DATE: _____

$$(A - \lambda I)x = 0$$

$$\begin{bmatrix} 5+6.033 & 9 & 3 \\ -3 & 5+6.033 & 6 \\ -1 & -5 & -3+6.033 \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} 0.7071 \\ -0.7071 \\ 0 \end{bmatrix}$$

for $\lambda = 0.767$

$$(A - \lambda I)x = 0$$

$$\begin{bmatrix} 5-0.767 & 9 & 3 \\ 3 & 5-0.767 & 6 \\ -1 & -5 & -3-0.767 \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} 0.5774 \\ 0.5774 \\ 0.5774 \end{bmatrix}$$

So, eigenvalues are $\lambda = -6.033$
and $\lambda = 0.767$.

eigen vectors $\begin{bmatrix} 0.7071 \\ -0.7071 \\ 0 \end{bmatrix}$,

$$\begin{bmatrix} 0.5774 \\ 0.5774 \\ 0.5774 \end{bmatrix}$$

DATE: _____

DAY: _____

5) What condition must a, b, c and d satisfy so that the following matrix are linearly independent.

$$M_{22} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Example # 20 Page # 241.

وَقُلْ رَبِّ زِدْنِي عِلْمًا

PUACP